

1. *Bubble Sort.*

A set of integers $\{1, 2, 3, \dots, n\}$ is to be sorted into ascending order from left to right.

- (a) State the initial ordering which gives the worst case. What is the time complexity in the worst case? [Define complexity to be the number of comparisons required.]

The best case for Bubble Sort has the integers in order to begin with.

- (b) How many comparisons are required on a list of length n , in the best case?

A 'near-best case' might have two adjacent integers swapped around, e.g. $\{1, 2, 3, 5, 4, 6, 7, 8\}$.

- (c) How many passes are needed to complete the algorithm in the near-best case?

- (d) What is the time complexity in the near-best case?

We might consider an average-case scenario for Bubble Sort to be one in which the number of passes required is roughly half the length of the list.

- (e) What is the average-case complexity of Bubble Sort?

2. *Shuttle sort.*

A set of integers $\{1, 2, 3, \dots, n\}$ is to be sorted into ascending order from left to right.

- (a) State the initial ordering which gives the worst case. What is the time complexity in the worst case?

- (b) How many comparisons are required on a list of length n , in the best case?

We might consider an average-case scenario for Shuttle Sort to be one in which we have to shuttle back along half the list (roughly) to complete the pass.

- (c) What is the average-case complexity of Shuttle Sort?